

Jenn Leung & Chloe Loewith

EXPERIMENT

GRANT ALLOCATED \$8,400

One-Sentence Description

Using multiplayer games as an interface for communicating with biological computational materials, and benchmarking Synthetic Bioengineered Intelligence (SBI).

Proposed Project

Jenn and Chloe are building a simulation platform for living neurons, contributing to benchmarking intelligence in Synthetic Biological Intelligence (SBI). SBI is an emerging field that fuses living biological matter with electronic hardware to create self-learning, biohybrid systems. Jenn and Chloe's experimental project sits within the discourse of "brain-on-a-chip" and organoid intelligence.

There is no standard to encoding information to neurons as of now. Jenn and Chloe see this as a design challenge requiring multifaceted approaches. As early collaborators with Cortical Labs—the company that released the first commercially available biological computer and which experimented with teaching neuron culture to play Pong—Jenn and Chloe are building on this emerging research field by bringing their hybrid philosophy and design expertise.

Jenn and Chloe will study the simulation protocols across different systems and apply appropriate time scales, rate limits, response rates and stimulation/discretization patterns, in order to formalize communication with living neurons.

Their goal for the RRF project comes in two folds:

1. Software Development. Building on their previous work on the benchmarking platform.
2. Hardware Research. Jenn and Chloe are diving into adaptive biological chips with living substrates. With RRF funding, they will do several experimental runs.

The core goal is to establish reproducible benchmarking standards at the API layer that allow researchers to measure what synthetic-biology hybridized systems are "doing" and optimizing their ability to "learn." Jenn and Chloe are building a principled, measurable protocol that standardizes both API and benchmarking for research in the field.

Revival Relevance

Jenn and Chloe are reviving an older research tradition (artificial life) with a new engineering toolkit for benchmarking. They are inspired by the "bottom-up" approach to artificial life (Alife), characterized by the emphasis on emergence and complex systems. As opposed to directly programming behaviors, they focus on constructing an environment for SBI, allowing researchers and designers to use reinforcement learning algorithms within the game environment to train neuronal culture as game agents.

This is a right moment as recent technology advances finally makes sustained, remote experimentation practical for this stream of work: better-vascularized, longer-living brain organoids being available; Cortical Labs launching cloud access to their hardware.

Additionally, this project is highly interdisciplinary, both in the methodologies Jenn and Chloe are integrating, and in the outputs and presentation formats they are using to share their work to a wide range of audiences. At the rigor level of their research question, it is hard to find funding for the illegible, multifaceted forms that emerge from their collaboration. Jenn and Chloe have been working across institutions without a dedicated funding home for this specific line of work. I believe that their combined backgrounds in design, art, bioethics and game development make them rare and strong entrants into the emerging SBI field, and the Revival Fund is uniquely suited to support their work.

Wendi's Notes

1. **Jenn and Chloe's combined background is well positioned** to tackle the design challenge of benchmarking SBI.

Jenn was the first creative at Cortical Labs, and Final Spark got in touch with them as well. Jenn is currently a researcher at the [Lifefabs institute](#) and recently [gave talks there](#) about her and Chloe's work. Chloe was at the Cambridge Leverhume Center of Future Intelligence researching bioethics and synthetic biology, and currently a Fellow at the Future Impact Group (Oxford x NYU) on embodiment for digital minds.

2. **Jenn and Chloe are uniquely positioned at the cutting edge of SBI, design and bioethics.**

Only two major companies are working on similar pursuits: Final Spark and Cortical Labs, and Jenn and Chloe are in touch with both. [Cortical Labs' CL1](#) is the world's first commercially available biological computer, powered by approximately 800,000 lab-grown human neurons. Jenn was the first artist-technologist to collaborate with Cortical Labs. She and Chloe have been using CL1 with special access to their machines since summer 2025,

They are also friends with Daniel Burger (also our grantee), who was the neuroengineer behind FinalSpark's cloud project, and they have been planning to collaborate on developing design strategies for brain-in-vat systems for artificial life research.

3. **Jenn and Chloe are self-driven, and have already made major progress while pursuing this inquiry with minimal funding.**

They met at the Antikythera studio program in 2024, and started collaborating then: they have produced two papers for the MIT Antikythera Journal and NeurIPS 2025, and are in the process of writing a philosophy paper with Laura Tripaldi (NYU Shanghai) and Brett Kagan (CSO, Cortical Labs). Jenn collaborated with the Michael Levin Lab and artist Agnieszka Kuran for Rhizome 7x7 and presented it at the New Museum. They will also be presenting at Gray Area and Artificial Life conference to do a tutorial session on signal benchmarking with Wes Clawson (Tufts).